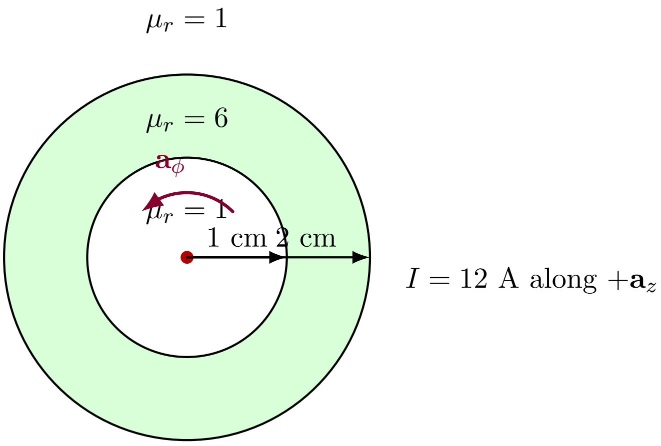
1. A conducting filament at $z = 0$ carries 12 A in the $\mathbf{a}_z$ direction. Let

$$\mu_r = \begin{cases} 1, & \rho < 1 \text{ cm}, \\ 6, & 1 < \rho < 2 \text{ cm}, \\ 1, & \rho > 2 \text{ cm}. \end{cases}$$

Find: (a) $\mathbf{H}$ everywhere, (b) $\mathbf{B}$ everywhere.

Solution



Symmetry

Because the current filament is along the z-axis and the material regions are concentric cylindrical shells, the field has cylindrical symmetry.

Hence,

$$\mathbf{H} = H_\phi(\rho)\,\mathbf{a}_\phi.$$

That is, $\mathbf{H}$ is purely circumferential.

Use Ampère's law

Choose a circular Amperian loop of radius ρ, centered on the z-axis. Then

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}.$$

Since $\mathbf{H}$ is tangential and constant in magnitude on the circular path,

$$H_\phi(\rho)\,(2\pi\rho) = 12.$$

Therefore,

$$H_\phi(\rho) = \frac{12}{2\pi\rho} = \frac{6}{\pi\rho}.$$

So,

$$\boxed{\mathbf{H}(\rho) = \frac{6}{\pi\rho}\,\mathbf{a}_\phi \quad \text{A/m}, \qquad \rho > 0}$$

This is the same in all three regions, because Ampère's law for $\mathbf{H}$ depends only on the enclosed *free current*, not on μ_r.

Magnetic flux density B

Now

$$\mathbf{B} = \mu\mathbf{H} = \mu_0\mu_r\mathbf{H}.$$

So in each region,

Region 1: $\rho < 1$ cm, $\mu_r = 1$

$$\mathbf{B} = \mu_0\left(\frac{6}{\pi\rho}\mathbf{a}_\phi\right)$$

hence

$$\boxed{\mathbf{B}(\rho) = \frac{6\mu_0}{\pi\rho}\,\mathbf{a}_\phi \quad \text{T}, \qquad \rho < 1 \text{ cm}}$$

Region 2: $1 < \rho < 2$ cm, $\mu_r = 6$

$$\mathbf{B} = 6\mu_0\left(\frac{6}{\pi\rho}\mathbf{a}_\phi\right)$$

so

$$\boxed{\mathbf{B}(\rho) = \frac{36\mu_0}{\pi\rho}\,\mathbf{a}_\phi \quad \text{T}, \qquad 1 < \rho < 2 \text{ cm}}$$

Region 3: $\rho > 2$ cm, $\mu_r = 1$

$$\mathbf{B} = \mu_0 \left(\frac{6}{\pi\rho} \mathbf{a}_\phi \right)$$

thus

$$\boxed{\mathbf{B}(\rho) = \frac{6\mu_0}{\pi\rho} \mathbf{a}_\phi \quad \text{T}, \qquad \rho > 2 \text{ cm}}$$

Final Answer

$$\boxed{\mathbf{H}(\rho) = \frac{6}{\pi\rho} \mathbf{a}_\phi \quad \text{A/m}, \qquad \rho > 0}$$

$$\boxed{\mathbf{B}(\rho) = \begin{cases} \dfrac{6\mu_0}{\pi\rho} \mathbf{a}_\phi, & \rho < 1 \text{ cm}, \\ \dfrac{36\mu_0}{\pi\rho} \mathbf{a}_\phi, & 1 < \rho < 2 \text{ cm}, \\ \dfrac{6\mu_0}{\pi\rho} \mathbf{a}_\phi, & \rho > 2 \text{ cm}. \end{cases}}$$

Optional numerical form

Using $\mu_0 = 4\pi \times 10^{-7}$ H/m,

$$\frac{6\mu_0}{\pi\rho} = \frac{24 \times 10^{-7}}{\rho} = \frac{2.4 \times 10^{-6}}{\rho},$$

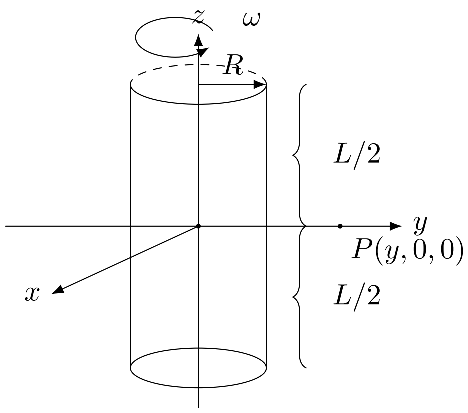
so

$$\mathbf{B}(\rho) = \begin{cases} \dfrac{2.4 \times 10^{-6}}{\rho} \mathbf{a}_\phi, & \rho < 1 \text{ cm}, \\ \dfrac{1.44 \times 10^{-5}}{\rho} \mathbf{a}_\phi, & 1 < \rho < 2 \text{ cm}, \\ \dfrac{2.4 \times 10^{-6}}{\rho} \mathbf{a}_\phi, & \rho > 2 \text{ cm}, \end{cases}$$

with ρ in meters.

---

2. A thin glass rod of radius R and length L carries a uniform surface charge density σ. It is set spinning about its axis with angular velocity ω. Find the magnetic field at a point in the xy-plane located at a distance $y \gg R$ from the axis. *(Hint: Treat the rod as a stack of magnetic dipoles.)* Full marks will be awarded only if the final answer is presented in a fully simplified form.



Solution

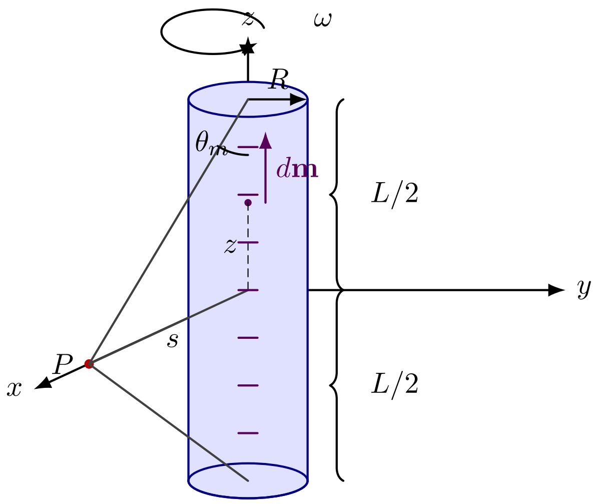


Figure 1: Spinning charged rod treated as a stack of magnetic dipoles. The observation point P is at a distance s from the axis, with $s \gg R$.

For a magnetic dipole $\mathbf{m} = m\,\hat{\mathbf{z}}$ at the origin, the magnetic field is

$$\mathbf{B} = \frac{\mu_0}{4\pi r^3}\left(2m\cos\theta\,\hat{\mathbf{r}} + m\sin\theta\,\hat{\boldsymbol{\theta}}\right). \tag{1}$$

Now, for the xz plane ($\phi = 0$),

$$\hat{\mathbf{r}} = \sin\theta\,\hat{\mathbf{x}} + \cos\theta\,\hat{\mathbf{z}}, \qquad \hat{\boldsymbol{\theta}} = \cos\theta\,\hat{\mathbf{x}} - \sin\theta\,\hat{\mathbf{z}}.$$

Therefore,

$$\mathbf{B} = \frac{\mu_0 m}{4\pi r^3}\left[2\cos\theta(\sin\theta\,\hat{\mathbf{x}} + \cos\theta\,\hat{\mathbf{z}}) + \sin\theta(\cos\theta\,\hat{\mathbf{x}} - \sin\theta\,\hat{\mathbf{z}})\right] \tag{2}$$

$$= \frac{\mu_0 m}{4\pi r^3}\left[(2\sin\theta\cos\theta + \sin\theta\cos\theta)\hat{\mathbf{x}} + (2\cos^2\theta - \sin^2\theta)\hat{\mathbf{z}}\right] \tag{3}$$

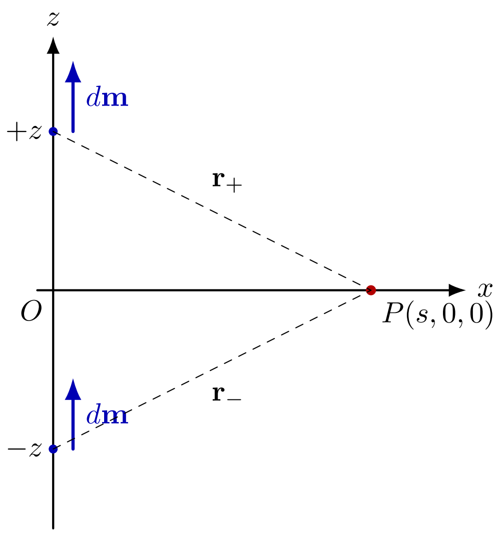
$$= \frac{\mu_0 m}{4\pi r^3}\left[3\sin\theta\cos\theta\,\hat{\mathbf{x}} + (2\cos^2\theta - \sin^2\theta)\hat{\mathbf{z}}\right]. \tag{4}$$

Since

$$2\cos^2\theta - \sin^2\theta = 3\cos^2\theta - 1,$$

we may also write

$$\mathbf{B} = \frac{\mu_0 m}{4\pi r^3}\left[3\sin\theta\cos\theta\,\hat{\mathbf{x}} + (3\cos^2\theta - 1)\hat{\mathbf{z}}\right]. \tag{5}$$



Step 2: Replace the spinning rod by a stack of dipoles

Take a thin charged strip of width dz on the cylindrical surface.
The charge on this strip is

$$dq = \sigma(2\pi R\,dz).$$

Since the rod rotates with angular speed ω, the equivalent current of the strip is

$$dI = \frac{dq}{T} = \frac{\sigma(2\pi R\,dz)}{2\pi/\omega} = \sigma\omega R\,dz.$$

Hence its magnetic dipole moment is

$$d\mathbf{m} = dI(\pi R^2)\,\hat{\mathbf{z}} = \pi\sigma\omega R^3\,dz\,\hat{\mathbf{z}}.$$

Let

$$\mathcal{M} \equiv \frac{dm}{dz} = \pi\sigma\omega R^3.$$

Choose the field point to be

$$P = (s, 0, 0),$$

that is, on the x-axis. By cylindrical symmetry, this is sufficient for any point in the equatorial plane at distance s from the rod axis.

Now consider two dipole elements, one at z and the other at $-z$, with the same dipole moment

$$d\mathbf{m} = dm\,\hat{\mathbf{z}}.$$

For the element at $-z$, let the polar angle at P be θ, so that

$$\sin\theta = \frac{s}{r}, \qquad \cos\theta = \frac{z}{r}, \qquad r = \sqrt{s^2 + z^2}.$$

Then for the element at $+z$, the corresponding polar angle is $\pi - \theta$. Therefore,

$$\sin(\pi - \theta) = \sin\theta, \qquad \cos(\pi - \theta) = -\cos\theta.$$

Using the dipole-field result from Step 1,

$$d\mathbf{B} = \frac{\mu_0\, dm}{4\pi r^3}\left[3\sin\theta\cos\theta\,\hat{\mathbf{x}} + (3\cos^2\theta - 1)\hat{\mathbf{z}}\right],$$

we get for the lower element $(-z)$:

$$d\mathbf{B}_- = \frac{\mu_0\, dm}{4\pi r^3}\left[3\sin\theta\cos\theta\,\hat{\mathbf{x}} + (3\cos^2\theta - 1)\hat{\mathbf{z}}\right],$$

and for the upper element $(+z)$:

$$d\mathbf{B}_+ = \frac{\mu_0\, dm}{4\pi r^3}\left[-3\sin\theta\cos\theta\,\hat{\mathbf{x}} + (3\cos^2\theta - 1)\hat{\mathbf{z}}\right].$$

Hence, when the two are added,

$$d\mathbf{B}_{\text{pair}} = d\mathbf{B}_+ + d\mathbf{B}_- = \frac{\mu_0\, dm}{2\pi r^3}(3\cos^2\theta - 1)\,\hat{\mathbf{z}}.$$

So the x-components cancel exactly, while the z-components add.

Step 3: Integrate over the rod

Now sum over all such symmetric pairs from $z = 0$ to $z = L/2$:

$$\mathbf{B} = \frac{\mu_0 \mathcal{M}}{2\pi} \int_0^{L/2} \frac{3\cos^2\theta - 1}{r^3}\, dz\, \hat{\mathbf{z}}.$$

Use

$$r = \frac{s}{\sin\theta}, \qquad z = s\cot\theta, \qquad dz = -s\csc^2\theta\, d\theta = -\frac{s}{\sin^2\theta}\, d\theta.$$

Also, when

$$z = 0, \qquad \theta = \frac{\pi}{2},$$

and when

$$z = \frac{L}{2}, \qquad \theta = \theta_m,$$

where

$$\tan\theta_m = \frac{s}{L/2} = \frac{2s}{L}.$$

Therefore,

$$\mathbf{B} = \frac{\mu_0 \mathcal{M}}{2\pi} \int_0^{L/2} \frac{3\cos^2\theta - 1}{r^3}\, dz\, \hat{\mathbf{z}} \tag{6}$$

$$= \frac{\mu_0 \mathcal{M}}{2\pi} \int_{\pi/2}^{\theta_m} (3\cos^2\theta - 1) \left(\frac{\sin^3\theta}{s^3}\right) \left(-\frac{s}{\sin^2\theta}\right) d\theta\, \hat{\mathbf{z}} \tag{7}$$

$$= -\frac{\mu_0 \mathcal{M}}{2\pi s^2} \int_{\pi/2}^{\theta_m} (3\cos^2\theta - 1)\sin\theta\, d\theta\, \hat{\mathbf{z}}. \tag{8}$$

Reverse the limits:

$$\mathbf{B} = \frac{\mu_0 \mathcal{M}}{2\pi s^2} \int_{\theta_m}^{\pi/2} (3\cos^2\theta - 1)\sin\theta\, d\theta\, \hat{\mathbf{z}}.$$

Now,

$$\int (3\cos^2\theta - 1)\sin\theta\, d\theta = -\cos^3\theta + \cos\theta.$$

So

$$\mathbf{B} = \frac{\mu_0 \mathcal{M}}{2\pi s^2} \left[-\cos^3\theta + \cos\theta\right]_{\theta_m}^{\pi/2} \hat{\mathbf{z}} \tag{9}$$

$$= \frac{\mu_0 \mathcal{M}}{2\pi s^2} \left(0 - (-\cos^3\theta_m + \cos\theta_m)\right) \hat{\mathbf{z}} \tag{10}$$

$$= \frac{\mu_0 \mathcal{M}}{2\pi s^2} \left(\cos^3\theta_m - \cos\theta_m\right) \hat{\mathbf{z}} \tag{11}$$

$$= -\frac{\mu_0 \mathcal{M}}{2\pi s^2} \cos\theta_m \sin^2\theta_m \, \hat{\mathbf{z}}. \tag{12}$$

Now,

$$\sin\theta_m = \frac{s}{\sqrt{s^2 + (L/2)^2}}, \qquad \cos\theta_m = \frac{L/2}{\sqrt{s^2 + (L/2)^2}}.$$

Substituting,

$$\mathbf{B} = -\frac{\mu_0 \mathcal{M}}{2\pi s^2} \left(\frac{L/2}{\sqrt{s^2 + (L/2)^2}}\right) \left(\frac{s^2}{s^2 + (L/2)^2}\right) \hat{\mathbf{z}}.$$

Hence,

$$\mathbf{B} = -\frac{\mu_0 \mathcal{M} \, L}{4\pi \left[s^2 + (L/2)^2\right]^{3/2}} \hat{\mathbf{z}}.$$

Finally, using

$$\mathcal{M} = \pi\sigma\omega R^3,$$

we get

$$\boxed{\mathbf{B} = -\frac{\mu_0 \sigma\omega R^3 L}{4\left[s^2 + (L/2)^2\right]^{3/2}} \hat{\mathbf{z}}}$$

If only the magnitude is required,

$$\boxed{B = \frac{\mu_0 \sigma\omega R^3 L}{4\left[s^2 + (L/2)^2\right]^{3/2}}}$$

Final Answer

$$\boxed{B = \frac{\mu_0 \sigma\omega R^3 L}{4\left[s^2 + (L/2)^2\right]^{3/2}}}$$

This is the **magnitude** of the magnetic field at the observation point. The actual direction is along the rod axis and is determined by the **right-hand rule** for the given sense of rotation.

3. An alternating current $I = I_0 \cos(\omega t)$ flows down a long straight wire, and returns along a coaxial conducting tube of radius a.

   (a) In what *direction* does the induced electric field point (radial, circumferential, or longitudinal)?

   (b) Assuming the field goes to zero as $s \to \infty$, find $\mathbf{E}(s,t)$.

Solution

The current flows along the axis of the system, so the magnetic field produced by the current is azimuthal (circumferential), i.e.

$$\mathbf{B} = B_\phi(s,t)\,\hat{\boldsymbol{\phi}}.$$

Because the geometry is infinitely long and cylindrically symmetric:

- nothing depends on z,
- nothing depends on ϕ,
- the induced electric field cannot have a circumferential component by symmetry,
- and it also cannot be radial in this case. (Apply Faraday's law)

So the induced electric field must be **longitudinal**, i.e. along the wire axis:

$$\boxed{\mathbf{E}(s,t) = E_z(s,t)\,\hat{\mathbf{z}}}$$

Thus, the answer to part (a) is:

$$\boxed{\text{The induced electric field is longitudinal.}}$$

Magnetic field due to the coaxial current

For a circular Amperian loop of radius s:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}.$$

Hence,

$$B_\phi(s,t) = \begin{cases} \dfrac{\mu_0 I(t)}{2\pi s}, & 0 < s < a, \\ 0, & s > a, \end{cases}$$

because for $s > a$ the forward current in the inner wire and the return current in the outer tube cancel, so the net enclosed current is zero.

Using

$$I(t) = I_0 \cos(\omega t),$$

we get

$$B_\phi(s,t) = \begin{cases} \dfrac{\mu_0 I_0 \cos(\omega t)}{2\pi s}, & 0 < s < a, \\ 0, & s > a. \end{cases}$$

Use Faraday's law

Faraday's law in differential form is

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

Since

$$\mathbf{E} = E_z(s,t)\,\hat{\mathbf{z}},$$

the only nonzero component of $\nabla \times \mathbf{E}$ is the ϕ-component:

$$(\nabla \times \mathbf{E})_\phi = -\frac{\partial E_z}{\partial s}.$$

Therefore,

$$-\frac{\partial E_z}{\partial s} = -\frac{\partial B_\phi}{\partial t},$$

so that

$$\frac{\partial E_z}{\partial s} = \frac{\partial B_\phi}{\partial t}.$$

For $0 < s < a$,

$$B_\phi(s,t) = \frac{\mu_0 I(t)}{2\pi s},$$

hence

$$\frac{\partial B_\phi}{\partial t} = \frac{\mu_0}{2\pi s}\frac{dI}{dt}.$$

So,

$$\frac{\partial E_z}{\partial s} = \frac{\mu_0}{2\pi s}\frac{dI}{dt}.$$

Integrate to obtain E_z

Integrate with respect to s:

$$E_z(s,t) = \frac{\mu_0}{2\pi}\frac{dI}{dt}\ln s + C(t).$$

To determine the constant, use the condition at $s > a$.

Field outside the coaxial tube

For $s > a$, we have

$$\mathbf{B} = 0.$$

Thus,

$$\nabla \times \mathbf{E} = 0.$$

By symmetry, the only possible field is again of the form

$$\mathbf{E} = E_z(s,t)\hat{\mathbf{z}}.$$

Then

$$-\frac{\partial E_z}{\partial s} = 0 \quad \Longrightarrow \quad E_z = \text{constant}.$$

But the problem states that the field goes to zero as $s \to \infty$, so this constant must be zero:

$$\boxed{\mathbf{E}(s,t) = 0, \qquad s > a.}$$

Since the tangential component of $\mathbf{E}$ must be continuous at $s = a$,

$$E_z(a,t) = 0.$$

Hence, for $0 < s < a$,

$$E_z(s,t) = \frac{\mu_0}{2\pi}\frac{dI}{dt}\ln\left(\frac{s}{a}\right).$$

This is equivalently written as

$$E_z(s,t) = -\frac{\mu_0}{2\pi}\frac{dI}{dt}\ln\left(\frac{a}{s}\right).$$

Now,

$$\frac{dI}{dt} = -\omega I_0 \sin(\omega t).$$

Therefore,

$$E_z(s,t) = \frac{\mu_0 \omega I_0}{2\pi} \ln\left(\frac{a}{s}\right) \sin(\omega t).$$

So the electric field is

$$\boxed{\mathbf{E}(s,t) = \frac{\mu_0 \omega I_0}{2\pi} \ln\left(\frac{a}{s}\right) \sin(\omega t)\, \hat{\mathbf{z}}, \qquad 0 < s < a}$$

and

$$\boxed{\mathbf{E}(s,t) = 0, \qquad s > a.}$$

Physical interpretation

The magnetic field between the conductors changes with time because the current is alternating. Faraday's law then requires an induced electric field. Since the magnetic field is azimuthal, the induced electric field must run *along* the axis.

Also note:

- when the current is decreasing, the induced electric field tends to support the original current direction,
- when the current is increasing, the induced electric field opposes the increase,

which is consistent with Lenz's law.

Final Answer

$$\boxed{\text{(a) The induced electric field is longitudinal.}}$$

$$\boxed{\mathbf{E}(s,t) = \begin{cases} \dfrac{\mu_0 \omega I_0}{2\pi} \ln\left(\dfrac{a}{s}\right) \sin(\omega t)\, \hat{\mathbf{z}}, & 0 < s < a, \\ 0, & s > a. \end{cases}}$$

4. According to quantum mechanics, the electron cloud for a hydrogen atom in the ground state has a charge density
$$\rho(r) = \frac{q}{\pi a^3} e^{-2r/a},$$
where q is the charge of the electron and a is the Bohr radius. Find the atomic polarizability of such an atom.

Hint: First calculate the electric field of the electron cloud, $\mathbf{E}_e(r)$; then expand the exponential, assuming $r \ll a$.

Solution

Physical idea

Suppose a uniform external electric field $\mathbf{E}$ is applied to the atom. Because the proton is positively charged and the electron cloud is negatively charged, the proton tends to shift slightly in the direction of the applied field, while the cloud remains approximately centered.

This small separation produces an induced dipole moment:
$$\mathbf{p} = \alpha \mathbf{E},$$
where α is the **atomic polarizability**.

So our task is:

- first find the electric field created by the electron cloud near the center,
- then determine the displacement of the proton in equilibrium,
- and finally read off α.

Charge enclosed within radius r

The electron cloud has charge density
$$\rho(r) = \frac{q}{\pi a^3} e^{-2r/a},$$
where $q < 0$ since it is the charge of the electron.

To find the field due to this cloud at a distance r from its center, we first compute the charge enclosed inside a sphere of radius r:

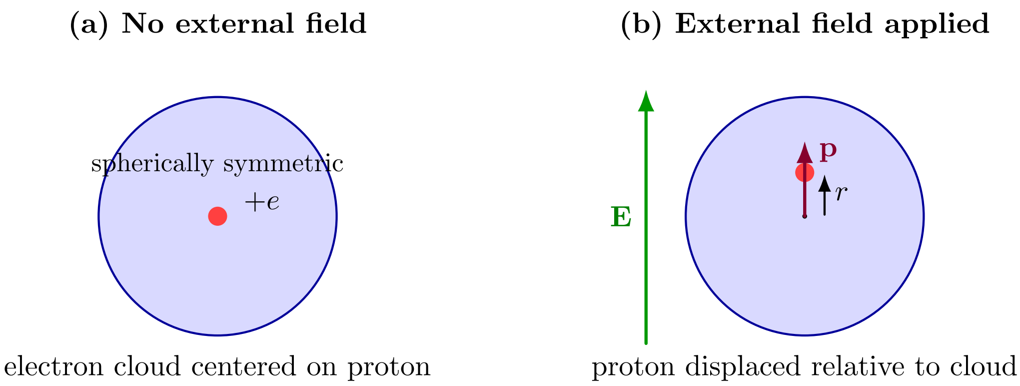


Figure 2: Hydrogen atom modeled as a spherically symmetric electron cloud and a proton. In an external electric field, the proton shifts slightly, producing an induced dipole moment.

$$Q_{\text{enc}}(r) = \int_0^r \rho(r')\, 4\pi r'^2 \, dr' \tag{13}$$

$$= \int_0^r \frac{q}{\pi a^3} e^{-2r'/a}\, 4\pi r'^2 \, dr' \tag{14}$$

$$= \frac{4q}{a^3} \int_0^r r'^2 e^{-2r'/a} \, dr'. \tag{15}$$

Carrying out the integration gives

$$Q_{\text{enc}}(r) = q \left[1 - e^{-2r/a} \left(1 + \frac{2r}{a} + \frac{2r^2}{a^2} \right) \right].$$

Electric field of the electron cloud

By spherical symmetry, Gauss's law gives

$$\mathbf{E}_e(r) = \frac{1}{4\pi\varepsilon_0} \frac{Q_{\text{enc}}(r)}{r^2} \, \hat{\mathbf{r}}.$$

Hence,

$$\mathbf{E}_e(r) = \frac{q}{4\pi\varepsilon_0 r^2} \left[1 - e^{-2r/a} \left(1 + \frac{2r}{a} + \frac{2r^2}{a^2} \right) \right] \hat{\mathbf{r}}.$$

This is the exact electric field due to the electron cloud.

Small-r approximation

We are interested in the field very near the center, because the displacement of the proton is small. So we use the approximation $r \ll a$.

Expand the exponential:

$$e^{-2r/a} = 1 - \frac{2r}{a} + \frac{2r^2}{a^2} - \frac{4r^3}{3a^3} + \cdots$$

Now multiply by

$$\left(1 + \frac{2r}{a} + \frac{2r^2}{a^2}\right):$$

$$e^{-2r/a}\left(1 + \frac{2r}{a} + \frac{2r^2}{a^2}\right) = 1 - \frac{4r^3}{3a^3} + \cdots$$

Therefore,

$$1 - e^{-2r/a}\left(1 + \frac{2r}{a} + \frac{2r^2}{a^2}\right) = \frac{4r^3}{3a^3}.$$

So the enclosed charge becomes

$$Q_{\text{enc}}(r) \approx q\,\frac{4r^3}{3a^3}.$$

Substituting into Gauss's law result,

$$\mathbf{E}_e(r) \approx \frac{1}{4\pi\varepsilon_0}\frac{1}{r^2}\left(q\frac{4r^3}{3a^3}\right)\hat{\mathbf{r}} \tag{16}$$

$$= \frac{q\,r}{3\pi\varepsilon_0 a^3}\hat{\mathbf{r}}. \tag{17}$$

Thus, near the center,

$$\boxed{\mathbf{E}_e(r) \approx \frac{q\,r}{3\pi\varepsilon_0 a^3}\,\hat{\mathbf{r}}}$$

Since $q < 0$, this field is directed toward the center, so it provides the restoring force.

Equilibrium in an external field

Let the external electric field be $\mathbf{E}$. If the proton is displaced by a small vector $\mathbf{r}$, then the force on the proton due to the electron cloud is restoring.

The proton has charge $+e$, and since $q = -e$, we can write the cloud field as

$$\mathbf{E}_e(\mathbf{r}) = -\frac{e}{3\pi\varepsilon_0 a^3}\mathbf{r}.$$

At equilibrium, the total force on the proton must vanish:

$$e\mathbf{E} + e\mathbf{E}_e = 0.$$

Hence,

$$\mathbf{E} + \mathbf{E}_e = 0.$$

So

$$\mathbf{E} = \frac{e}{3\pi\varepsilon_0 a^3}\mathbf{r}.$$

Therefore,

$$\mathbf{r} = \frac{3\pi\varepsilon_0 a^3}{e}\,\mathbf{E}.$$

Induced dipole moment

The induced dipole moment is

$$\mathbf{p} = e\,\mathbf{r}.$$

Substitute the expression for $\mathbf{r}$:

$$\mathbf{p} = e\left(\frac{3\pi\varepsilon_0 a^3}{e}\mathbf{E}\right) = 3\pi\varepsilon_0 a^3\,\mathbf{E}.$$

Comparing with

$$\mathbf{p} = \alpha\mathbf{E},$$

we obtain

$$\boxed{\alpha = 3\pi\varepsilon_0 a^3.}$$

Final Answer

For the hydrogen atom with electron-cloud charge density

$$\rho(r) = \frac{q}{\pi a^3}e^{-2r/a},$$

the atomic polarizability is

$$\boxed{\alpha = 3\pi\varepsilon_0 a^3.}$$

---

5. The space between the plates of a parallel-plate capacitor is filled with two slabs of linear dielectric material. Each slab has thickness a, so the total distance between the plates is $2a$. Slab 1 has dielectric constant 2, and slab 2 has dielectric constant 1.5. The free charge density on the top plate is σ and on the bottom plate is $-\sigma$.

(a) Find the electric displacement $\mathbf{D}$ in each slab.

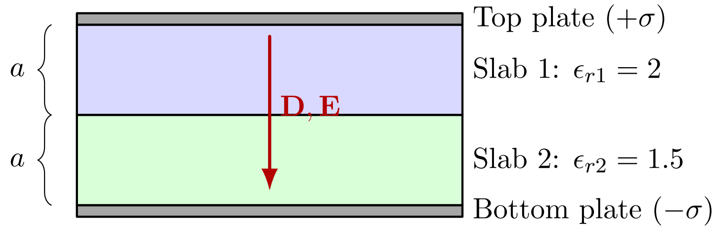
(b) Find the electric field $\mathbf{E}$ in each slab.

(c) Find the polarization $\mathbf{P}$ in each slab.

(d) Find the potential difference between the plates.

(e) Find the location and amount of all bound charge.

(f) Now that you know all the charge (free and bound), recalculate the field in each slab, and confirm your answer to (b).



Solution

Electric displacement D in each slab

Since there is no free charge at the interface between the two dielectric slabs, the normal component of $\mathbf{D}$ is continuous across the interface. Also, at the plates,

$$\mathbf{D} \cdot \hat{\mathbf{n}} = \sigma_{\text{free}}.$$

Hence the displacement field is the same in both slabs and equals the free surface charge density in magnitude:

$$\boxed{\mathbf{D}_1 = \mathbf{D}_2 = -\sigma\,\hat{\mathbf{z}}}$$

The minus sign indicates that the field points downward, from the positively charged top plate to the negatively charged bottom plate.

Electric field E in each slab

Using

$$\mathbf{D} = \epsilon\mathbf{E} = \epsilon_0\epsilon_r\mathbf{E},$$

we get, for slab 1 ($\epsilon_{r1} = 2$),

$$\mathbf{E}_1 = \frac{\mathbf{D}_1}{\epsilon_0\epsilon_{r1}} = -\frac{\sigma}{2\epsilon_0}\hat{\mathbf{z}}.$$

So,

$$\boxed{\mathbf{E}_1 = -\frac{\sigma}{2\epsilon_0}\hat{\mathbf{z}}}$$

For slab 2 ($\epsilon_{r2} = 1.5 = 3/2$),

$$\mathbf{E}_2 = \frac{\mathbf{D}_2}{\epsilon_0 \epsilon_{r2}} = -\frac{\sigma}{(3/2)\epsilon_0}\hat{\mathbf{z}} = -\frac{2\sigma}{3\epsilon_0}\hat{\mathbf{z}}.$$

Hence,

$$\boxed{\mathbf{E}_2 = -\frac{2\sigma}{3\epsilon_0}\hat{\mathbf{z}}}$$

Polarization P in each slab

Now use

$$\mathbf{P} = \mathbf{D} - \epsilon_0 \mathbf{E}.$$

From the earlier parts,

$$\mathbf{D}_1 = \mathbf{D}_2 = -\sigma\,\hat{\mathbf{z}},$$

$$\mathbf{E}_1 = -\frac{\sigma}{2\epsilon_0}\hat{\mathbf{z}}, \qquad \mathbf{E}_2 = -\frac{2\sigma}{3\epsilon_0}\hat{\mathbf{z}}.$$

Slab 1:

$$\mathbf{P}_1 = \mathbf{D}_1 - \epsilon_0 \mathbf{E}_1 \tag{18}$$

$$= -\sigma\,\hat{\mathbf{z}} - \epsilon_0\left(-\frac{\sigma}{2\epsilon_0}\hat{\mathbf{z}}\right) \tag{19}$$

$$= -\sigma\,\hat{\mathbf{z}} + \frac{\sigma}{2}\hat{\mathbf{z}} \tag{20}$$

$$= -\frac{\sigma}{2}\hat{\mathbf{z}}. \tag{21}$$

Hence,

$$\boxed{\mathbf{P}_1 = -\frac{\sigma}{2}\hat{\mathbf{z}}}$$

Slab 2:

$$\mathbf{P}_2 = \mathbf{D}_2 - \epsilon_0 \mathbf{E}_2 \tag{22}$$

$$= -\sigma\,\hat{\mathbf{z}} - \epsilon_0\left(-\frac{2\sigma}{3\epsilon_0}\hat{\mathbf{z}}\right) \tag{23}$$

$$= -\sigma\,\hat{\mathbf{z}} + \frac{2\sigma}{3}\hat{\mathbf{z}} \tag{24}$$

$$= -\frac{\sigma}{3}\hat{\mathbf{z}}. \tag{25}$$

Therefore,

$$\boxed{\mathbf{P}_2 = -\frac{\sigma}{3}\hat{\mathbf{z}}}$$

Potential difference between the plates

Each slab has thickness a, so the total potential difference is

$$V_{\text{top}} - V_{\text{bottom}} = -\int_{\text{bottom}}^{\text{top}} \mathbf{E} \cdot d\mathbf{l}.$$

Since the field is uniform in each slab,

$$V_{\text{top}} - V_{\text{bottom}} = -\left(E_{2z}a + E_{1z}a\right).$$

Substituting,

$$V_{\text{top}} - V_{\text{bottom}} = -\left(-\frac{2\sigma}{3\epsilon_0}a - \frac{\sigma}{2\epsilon_0}a\right) = \frac{7\sigma a}{6\epsilon_0}.$$

Therefore,

$$\boxed{V_{\text{top}} - V_{\text{bottom}} = \frac{7\sigma a}{6\epsilon_0}}$$

Bound charges: claculation

We now determine the bound charges directly from the jump in the electric field across each charged surface.

For any surface carrying total surface charge density σ_s,

$$E_{z,\text{above}} - E_{z,\text{below}} = \frac{\sigma_s}{\epsilon_0}.$$

Top boundary (top plate / slab 1): Above the boundary (inside the conductor), $E_z = 0$. Below the boundary (inside slab 1),

$$E_{1z} = -\frac{\sigma}{2\epsilon_0}.$$

So the total surface charge at the top boundary is

$$\sigma_{s,\text{top}} = \epsilon_0\left(0 - \left(-\frac{\sigma}{2\epsilon_0}\right)\right) = \frac{\sigma}{2}.$$

But this total charge consists of:

$$\sigma_{s,\text{top}} = \sigma + \sigma_{b,\text{top of slab 1}}.$$

Hence,

$$\sigma + \sigma_{b,\text{top of slab 1}} = \frac{\sigma}{2}$$

which gives

$$\boxed{\sigma_{b,\text{top of slab 1}} = -\frac{\sigma}{2}}$$

Bottom boundary (slab 2 / bottom plate): Above the boundary (inside slab 2),

$$E_{2z} = -\frac{2\sigma}{3\epsilon_0},$$

and below the boundary (inside the conductor), $E_z = 0$. Therefore,

$$\sigma_{s,\text{bottom}} = \epsilon_0 \left(-\frac{2\sigma}{3\epsilon_0} - 0 \right) = -\frac{2\sigma}{3}.$$

But this total charge is

$$\sigma_{s,\text{bottom}} = -\sigma + \sigma_{b,\text{bottom of slab 2}}.$$

So,

$$-\sigma + \sigma_{b,\text{bottom of slab 2}} = -\frac{2\sigma}{3},$$

hence

$$\boxed{\sigma_{b,\text{bottom of slab 2}} = +\frac{\sigma}{3}}$$

Interface between the slabs: The net surface charge at the interface is obtained from the jump in E_z:

$$\sigma_{s,\text{interface}} = \epsilon_0 \left(E_{1z} - E_{2z} \right).$$

Substituting,

$$\sigma_{s,\text{interface}} = \epsilon_0 \left(-\frac{\sigma}{2\epsilon_0} + \frac{2\sigma}{3\epsilon_0} \right) = \frac{\sigma}{6}.$$

Therefore,

$$\boxed{\sigma_{s,\text{interface}} = \frac{\sigma}{6}}$$

Now, since there is no volume charge in either slab, the total bound charge on each slab must be zero. Hence the two faces of each slab carry equal and opposite bound charge.

So for slab 1,

$$\boxed{\sigma_{b,\text{bottom of slab 1}} = +\frac{\sigma}{2}}$$

and for slab 2,

$$\boxed{\sigma_{b,\text{top of slab 2}} = -\frac{\sigma}{3}}$$

Indeed, the interface net charge is

$$\left(+\frac{\sigma}{2}\right) + \left(-\frac{\sigma}{3}\right) = \frac{\sigma}{6},$$

as obtained above.

(Recalculate the field from all sheets

Now the sheets are:

$$\begin{aligned} &+\sigma \quad \text{(top plate)}, && -\frac{\sigma}{2} \quad \text{(top of slab 1)}, \\ &+\frac{\sigma}{2} \quad \text{(bottom of slab 1)}, && -\frac{\sigma}{3} \quad \text{(top of slab 2)}, \\ &+\frac{\sigma}{3} \quad \text{(bottom of slab 2)}, && -\sigma \quad \text{(bottom plate)}. \end{aligned}$$

For an infinite charged sheet of density σ_s, the field on either side has magnitude

$$\frac{|\sigma_s|}{2\epsilon_0},$$

directed away from the sheet if $\sigma_s > 0$, and toward the sheet if $\sigma_s < 0$.

Field in slab 1: Adding the contributions of all sheets gives

$$\boxed{\mathbf{E}_1 = -\frac{\sigma}{2\epsilon_0}\hat{\mathbf{z}}}$$

Field in slab 2: Similarly,

$$\boxed{\mathbf{E}_2 = -\frac{2\sigma}{3\epsilon_0}\hat{\mathbf{z}}}$$

Thus the result agrees exactly with part (b).

Final Answer

$$\boxed{\mathbf{D}_1 = \mathbf{D}_2 = -\sigma\,\hat{\mathbf{z}}}$$

$$\boxed{\mathbf{E}_1 = -\frac{\sigma}{2\epsilon_0}\hat{\mathbf{z}}, \qquad \mathbf{E}_2 = -\frac{2\sigma}{3\epsilon_0}\hat{\mathbf{z}}}$$

$$\boxed{V_{\text{top}} - V_{\text{bottom}} = \frac{7\sigma a}{6\epsilon_0}}$$

Bound charges:

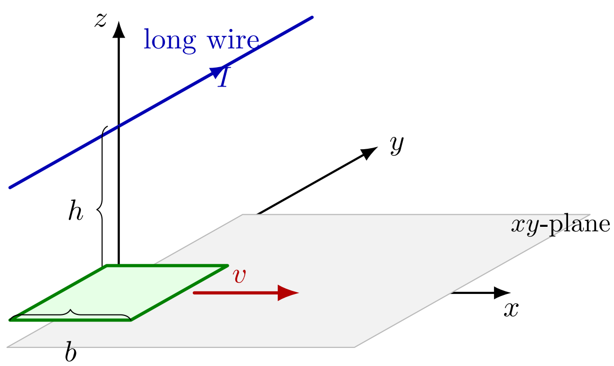
$$\boxed{\text{top of slab 1:} \;\; -\frac{\sigma}{2}, \quad \text{bottom of slab 1:} \;\; +\frac{\sigma}{2},}$$

$$\boxed{\text{top of slab 2:} \;\; -\frac{\sigma}{3}, \quad \text{bottom of slab 2:} \;\; +\frac{\sigma}{3}}$$

Interface net bound charge:

$$\boxed{\frac{\sigma}{6}}$$

6. A long straight stationary wire is parallel to the y-axis and passes through the point $(x, z) = (0, h)$; that is, it lies along the line $x = 0$, $z = h$. A current I flows in this wire, returning by a remote conductor whose field may be neglected. A square loop of side b lies in the xy-plane. Two of its sides are parallel to the long wire. The loop slides with constant speed v in the $+\hat{\mathbf{x}}$ direction. Find the magnitude of the electromotive force induced in the loop at the instant when the center of the loop crosses the y-axis.



Solution

Magnetic field due to the long wire

The wire runs along the y-direction and is located at $x = 0,\ z = h$. At a general point $(x, y, 0)$ of the loop, the perpendicular displacement from the wire is

$$\boldsymbol{\rho} = x\,\hat{\mathbf{x}} - h\,\hat{\mathbf{z}}, \qquad \rho = \sqrt{x^2 + h^2}.$$

For an infinite straight wire parallel to $\hat{\mathbf{y}}$,

$$\mathbf{B} = \frac{\mu_0 I}{2\pi\rho^2}\,\hat{\mathbf{y}} \times \boldsymbol{\rho}.$$

Now,

$$\hat{\mathbf{y}} \times \boldsymbol{\rho} = \hat{\mathbf{y}} \times (x\hat{\mathbf{x}} - h\hat{\mathbf{z}}) = -x\,\hat{\mathbf{z}} - h\,\hat{\mathbf{x}}.$$

Hence

$$\boxed{\mathbf{B}(x, y, 0) = \frac{\mu_0 I}{2\pi(x^2 + h^2)}\,(-h\,\hat{\mathbf{x}} - x\,\hat{\mathbf{z}})}$$

Since the loop lies in the xy-plane, the magnetic flux through it depends only on the component normal to the plane, namely B_z:

$$\boxed{B_z(x) = -\frac{\mu_0 I}{2\pi}\frac{x}{x^2 + h^2}}$$

Notice that B_z depends only on x, not on y.

Magnetic flux through the square loop

Let the x-coordinate of the loop center be x_c. Then the square spans

$$x \in \left[x_c - \frac{b}{2},\ x_c + \frac{b}{2}\right], \qquad y \in \left[-\frac{b}{2},\ \frac{b}{2}\right].$$

Therefore the flux through the loop is

$$\Phi(x_c) = \iint \mathbf{B} \cdot d\mathbf{a} = \int_{x_c - b/2}^{x_c + b/2} \int_{-b/2}^{b/2} B_z(x)\,dy\,dx \tag{26}$$

$$= b\int_{x_c - b/2}^{x_c + b/2} \left(-\frac{\mu_0 I}{2\pi}\frac{x}{x^2 + h^2}\right) dx. \tag{27}$$

So,

$$\Phi(x_c) = -\frac{\mu_0 I b}{2\pi} \int_{x_c-b/2}^{x_c+b/2} \frac{x}{x^2+h^2}\, dx.$$

Using

$$\int \frac{x}{x^2+h^2}\, dx = \frac{1}{2}\ln\left(x^2+h^2\right),$$

we get

$$\Phi(x_c) = -\frac{\mu_0 I b}{4\pi} \ln\left[\frac{(x_c+b/2)^2+h^2}{(x_c-b/2)^2+h^2}\right].$$

Thus,

$$\boxed{\Phi(x_c) = -\frac{\mu_0 I b}{4\pi} \ln\left[\frac{(x_c+b/2)^2+h^2}{(x_c-b/2)^2+h^2}\right]}$$

Induced emf

The loop slides with speed v in the $+\hat{\mathbf{x}}$ direction, so

$$x_c = vt \qquad \Rightarrow \qquad \frac{dx_c}{dt} = v.$$

By Faraday's law,

$$\mathcal{E} = -\frac{d\Phi}{dt} = -v\,\frac{d\Phi}{dx_c}.$$

Differentiate the flux:

$$\frac{d\Phi}{dx_c} = -\frac{\mu_0 I b}{2\pi}\left[\frac{x_c+b/2}{(x_c+b/2)^2+h^2} - \frac{x_c-b/2}{(x_c-b/2)^2+h^2}\right].$$

Hence

$$\mathcal{E} = \frac{\mu_0 I b v}{2\pi}\left[\frac{x_c+b/2}{(x_c+b/2)^2+h^2} - \frac{x_c-b/2}{(x_c-b/2)^2+h^2}\right].$$

Evaluate when the center crosses the y-axis

At the instant the center crosses the y-axis,

$$x_c = 0.$$

Therefore

$$\mathcal{E} = \frac{\mu_0 I b v}{2\pi}\left[\frac{b/2}{h^2+b^2/4} - \frac{-b/2}{h^2+b^2/4}\right] \tag{28}$$

$$= \frac{\mu_0 I b v}{2\pi}\left[\frac{b}{h^2+b^2/4}\right]. \tag{29}$$

So the magnitude of the induced emf is

$$\boxed{|\mathcal{E}| = \frac{\mu_0 I\, b^2\, v}{2\pi\left(h^2 + \dfrac{b^2}{4}\right)}}$$

An equivalent simplified form is

$$\boxed{|\mathcal{E}| = \frac{2\mu_0 I\, b^2 v}{\pi(4h^2+b^2)}}$$

Final Answer

At the instant when the center of the square loop crosses the y-axis, the induced emf has magnitude

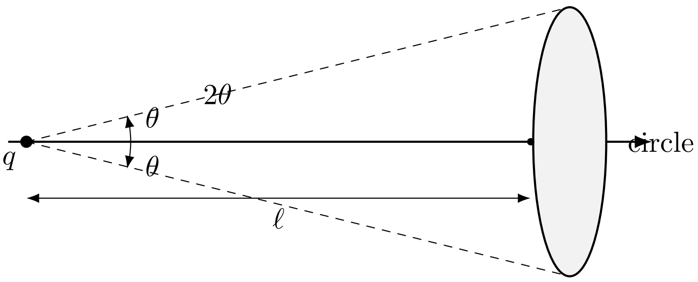
$$\boxed{|\mathcal{E}| = \frac{\mu_0 I\, b^2\, v}{2\pi\left(h^2 + \dfrac{b^2}{4}\right)} = \frac{2\mu_0 I\, b^2 v}{\pi(4h^2+b^2)}}$$

7. A point charge q is located at the origin. Consider the electric field flux through a circle a distance ℓ from q, subtending an angle 2θ, as shown below. Since there are no charges except at the origin, any surface that is bounded by the circle and that stays to the right of the origin must contain the same flux. (Why?)

   Calculate this flux by taking the surface to be:

   (a) the flat disk bounded by the circle;

   (b) the spherical cap (with the sphere centered at the origin) bounded by the circle.



Solution

Why the flux is the same for any surface bounded by the same circle

Suppose S_1 and S_2 are two different surfaces bounded by the same circle, both lying to the right of the origin. If we join them together (with opposite orientations), they form a closed surface.

Since this closed surface does *not* enclose the point charge at the origin, Gauss's law gives

$$\oint \mathbf{E} \cdot d\mathbf{a} = 0.$$

Therefore, the flux through S_1 must equal the flux through S_2.

So it is enough to compute the flux using whichever surface is easiest.

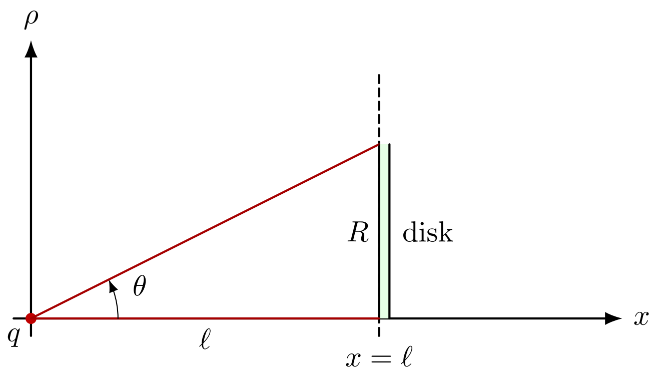
Key Idea

Any two surfaces sharing the same boundary circle and not enclosing the charge must have the same electric flux.

(a) Flux through the flat disk

Let the disk lie in the plane $x = \ell$, with center at $(\ell, 0, 0)$. Since the circle subtends an angle 2θ, its radius is

$$R = \ell \tan\theta.$$



Field and area element: Use cylindrical coordinates on the disk. A point on the disk is at distance

$$r = \sqrt{\ell^2 + \rho^2}$$

from the origin, where ρ is the radial coordinate measured from the center of the disk.

The electric field due to the point charge is

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}.$$

The area element on the disk is

$$dA = \rho\, d\rho\, d\phi,$$

and the angle between $\mathbf{E}$ and the normal to the disk is α, with

$$\cos\alpha = \frac{\ell}{r}.$$

Hence

$$d\Phi = \mathbf{E}\cdot d\mathbf{a} = E\cos\alpha\, dA = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\cdot\frac{\ell}{r}\,\rho\, d\rho\, d\phi.$$

So

$$d\Phi = \frac{q\ell}{4\pi\epsilon_0}\frac{\rho\, d\rho\, d\phi}{(\ell^2 + \rho^2)^{3/2}}.$$

Now integrate over the disk:

$$\Phi = \int_0^{2\pi}\int_0^R \frac{q\ell}{4\pi\epsilon_0}\frac{\rho\, d\rho\, d\phi}{(\ell^2 + \rho^2)^{3/2}}.$$

First integrate over ϕ:

$$\Phi = \frac{q\ell}{2\epsilon_0}\int_0^R \frac{\rho\, d\rho}{(\ell^2 + \rho^2)^{3/2}}.$$

Now,

$$\int \frac{\rho\, d\rho}{(\ell^2+\rho^2)^{3/2}} = -\frac{1}{\sqrt{\ell^2+\rho^2}}.$$

Therefore,

$$\Phi = \frac{q\ell}{2\epsilon_0}\left[-\frac{1}{\sqrt{\ell^2+\rho^2}}\right]_0^R \tag{30}$$

$$= \frac{q\ell}{2\epsilon_0}\left(\frac{1}{\ell} - \frac{1}{\sqrt{\ell^2+R^2}}\right). \tag{31}$$

Since

$$R = \ell\tan\theta,$$

we get

$$\sqrt{\ell^2+R^2} = \ell\sqrt{1+\tan^2\theta} = \ell\sec\theta.$$

Hence

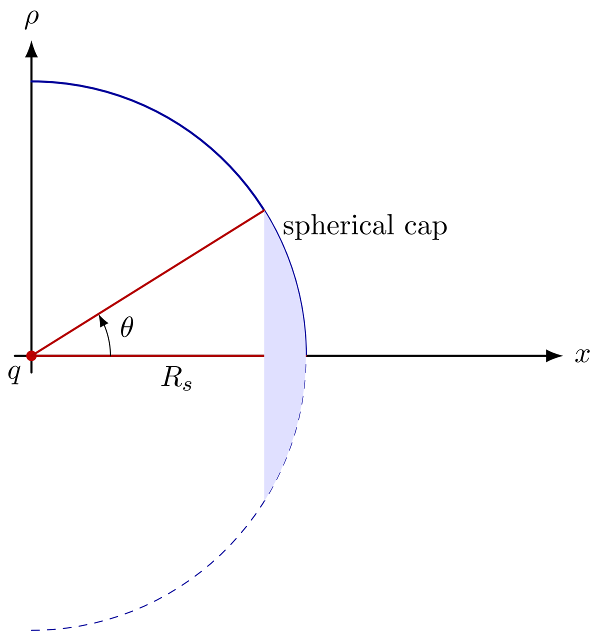
$$\frac{\ell}{\sqrt{\ell^2+R^2}} = \cos\theta.$$

So the flux becomes

$$\boxed{\Phi = \frac{q}{2\epsilon_0}(1-\cos\theta)}$$

(b) Flux through the spherical cap

Now take as surface the spherical cap whose boundary is the same circle, with the sphere centered at the origin.



Let the sphere have radius R_s. On this spherical surface, the electric field is everywhere radial and has constant magnitude

$$E = \frac{1}{4\pi\epsilon_0}\frac{q}{R_s^2}.$$

Also, the area vector $d\mathbf{a}$ is radial, so $\mathbf{E}$ is parallel to $d\mathbf{a}$. Therefore,

$$\mathbf{E} \cdot d\mathbf{a} = E\,dA.$$

So the flux through the cap is simply

$$\Phi = E \times (\text{area of spherical cap}).$$

The area of a spherical cap of half-angle θ is

$$A_{\text{cap}} = 2\pi R_s^2(1 - \cos\theta).$$

Hence

$$\Phi = \frac{1}{4\pi\epsilon_0}\frac{q}{R_s^2} \cdot 2\pi R_s^2(1 - \cos\theta) \tag{32}$$

$$= \boxed{\frac{q}{2\epsilon_0}(1 - \cos\theta)} \tag{33}$$

So the same result is obtained again.

Remark: relation to solid angle

This result can also be written in terms of the solid angle Ω subtended by the circle at the charge:

$$\Omega = 2\pi(1 - \cos\theta).$$

Then

$$\Phi = \frac{q}{4\pi\epsilon_0}\,\Omega.$$

This is the general flux formula for a point charge through any open surface.

Final Flux

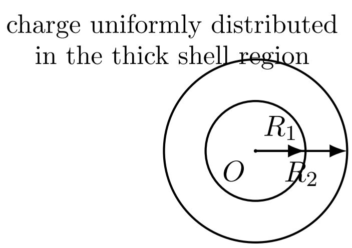
For both choices of surface,

$$\boxed{\Phi = \frac{q}{2\epsilon_0}(1 - \cos\theta)}$$

---

8. (a) A spherical shell with charge Q uniformly distributed throughout its volume has inner radius R_1 and outer radius R_2. Calculate (and make a rough plot of) the electric field as a function of r, for $0 < r < \infty$.

   (b) What is the potential at the center of the shell? You can let $R_2 = 2R_1$ in this part of the problem, to keep things from getting too messy. Give your answer in terms of $R \equiv R_1$.



Solution

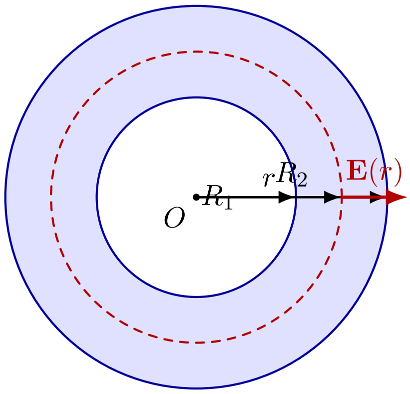
(a) Electric field everywhere

Because the charge distribution is spherically symmetric, the electric field must be purely radial:

$$\mathbf{E}(r) = E(r)\,\hat{\mathbf{r}}.$$

Let the uniform volume charge density be ρ. Since charge is distributed only in the region $R_1 < r < R_2$,

$$\rho = \frac{Q}{\frac{4}{3}\pi(R_2^3 - R_1^3)} = \frac{3Q}{4\pi(R_2^3 - R_1^3)}.$$



thick shell with a Gaussian sphere in the charged region

Case 1: $0 < r < R_1$ A spherical Gaussian surface of radius $r < R_1$ encloses no charge, so Gauss's law gives

$$E(4\pi r^2) = \frac{Q_{\text{enc}}}{\epsilon_0} = 0.$$

Hence,

$$\boxed{\mathbf{E} = 0, \qquad 0 < r < R_1.}$$

Case 2: $R_1 < r < R_2$ Now the enclosed charge is the charge contained between radii R_1 and r:

$$Q_{\text{enc}} = \rho\left(\frac{4}{3}\pi(r^3 - R_1^3)\right).$$

Using Gauss's law,

$$E(4\pi r^2) = \frac{Q_{\text{enc}}}{\epsilon_0} = \frac{1}{\epsilon_0}\rho\left(\frac{4}{3}\pi(r^3 - R_1^3)\right).$$

So

$$E(r) = \frac{1}{4\pi\epsilon_0}\frac{Q_{\text{enc}}}{r^2} \tag{34}$$
$$= \frac{1}{4\pi\epsilon_0}\frac{Q(r^3 - R_1^3)}{(R_2^3 - R_1^3)r^2}. \tag{35}$$

Therefore,

$$\boxed{\mathbf{E}(r) = \frac{Q}{4\pi\epsilon_0(R_2^3 - R_1^3)}\left(r - \frac{R_1^3}{r^2}\right)\hat{\mathbf{r}}, \qquad R_1 < r < R_2.}$$

Case 3: $r > R_2$ The entire charge Q is enclosed, so the shell behaves like a point charge at the center:

$$\boxed{\mathbf{E}(r) = \frac{Q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}}, \qquad r > R_2.}$$

Combining all three cases,

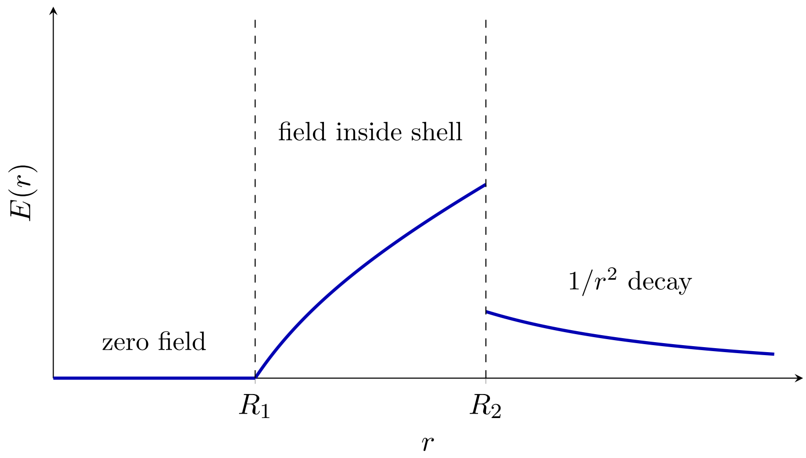
Electric Field

$$\boxed{\mathbf{E}(r) = \begin{cases} 0, & 0 < r < R_1, \\ \dfrac{Q}{4\pi\epsilon_0(R_2^3 - R_1^3)}\left(r - \dfrac{R_1^3}{r^2}\right)\hat{\mathbf{r}}, & R_1 < r < R_2, \\ \dfrac{Q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}}, & r > R_2. \end{cases}}$$

Rough plot of $E(r)$

For $Q > 0$, the magnitude behaves as follows:

- $E = 0$ for $0 < r < R_1$,
- it then rises smoothly through the charged region $R_1 < r < R_2$,
- and for $r > R_2$ it falls like $1/r^2$.



(b) Potential at the center

Because of spherical symmetry, the electric field is zero everywhere in the cavity $r < R_1$. Therefore the potential is constant throughout the cavity, so the potential at the center equals the potential anywhere inside $r < R_1$.

It is easiest to compute $V(0)$ by adding the contributions of thin spherical shells.

Take a thin shell of radius s and thickness ds. Its charge is

$$dq = \rho(4\pi s^2 ds).$$

The potential at the center due to this thin shell is

$$dV = \frac{1}{4\pi\epsilon_0}\frac{dq}{s}.$$

So,

$$dV = \frac{1}{4\pi\epsilon_0}\frac{\rho 4\pi s^2 ds}{s} \tag{36}$$

$$= \frac{\rho s}{\epsilon_0}\, ds. \tag{37}$$

Integrating from R_1 to R_2,

$$V(0) = \frac{\rho}{\epsilon_0}\int_{R_1}^{R_2} s\, ds = \frac{\rho}{2\epsilon_0}(R_2^2 - R_1^2).$$

Substitute

$$\rho = \frac{3Q}{4\pi(R_2^3 - R_1^3)}:$$

$$V(0) = \frac{3Q(R_2^2 - R_1^2)}{8\pi\epsilon_0(R_2^3 - R_1^3)}.$$

Now let $R_2 = 2R_1$, and define $R \equiv R_1$. Then

$$R_2 = 2R, \qquad R_2^2 - R_1^2 = 4R^2 - R^2 = 3R^2,$$
$$R_2^3 - R_1^3 = 8R^3 - R^3 = 7R^3.$$

Hence,

$$V(0) = \frac{3Q(3R^2)}{8\pi\epsilon_0(7R^3)} \tag{38}$$

$$= \boxed{\frac{9Q}{56\pi\epsilon_0 R}}. \tag{39}$$

Final Answers

$$\boxed{\mathbf{E}(r) = \begin{cases} 0, & 0 < r < R_1, \\ \dfrac{Q}{4\pi\epsilon_0(R_2^3 - R_1^3)}\left(r - \dfrac{R_1^3}{r^2}\right)\hat{\mathbf{r}}, & R_1 < r < R_2, \\ \dfrac{Q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}}, & r > R_2, \end{cases}}$$

$$\boxed{V(0) = \frac{9Q}{56\pi\epsilon_0 R} \quad \text{for } R_2 = 2R,\ R = R_1.}$$

9. A square conducting loop is cut out of a thick aluminum sheet, so the loop has a uniform cross-sectional area everywhere. It is placed in a vertical plane and allowed to fall downward under gravity. The upper part of the loop lies inside a region of uniform magnetic field $\mathbf{B}$, directed *into the page*, while the lower part lies outside the field region, as shown in the figure. As the loop falls, the magnetic flux through it changes, inducing a current in the loop. Assume the following idealizations:

(1)the magnetic field is uniform inside the shaded region and zero outside, (2) air resistance is neglected, (3) self-inductance of the loop is neglected, (4) aluminum has resistivity ρ and mass density ρ_m. If the magnetic field strength is $B = 1$ T, Take typical values for aluminum:

$$\rho \approx 2.7 \times 10^{-8}\ \Omega \cdot \text{m}, \qquad \rho_m \approx 2.70 \times 10^3\ \text{kg/m}^3,$$

answer the following:

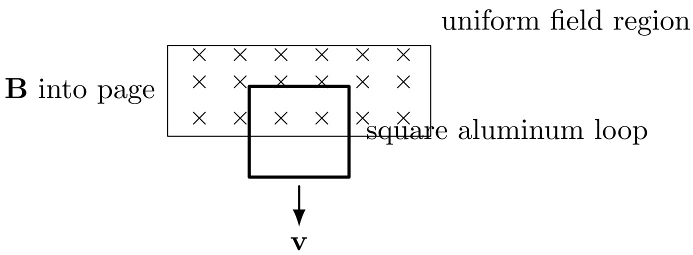
(a) Find the terminal velocity of the loop.

(b) Find the velocity as a function of time, $v(t)$, assuming it is released from rest.

(c) How long does it take to reach 90% of the terminal velocity?

(d) Show explicitly that the dimensions of the loop cancel out, and then evaluate the result numerically in SI units.

(e) What would happen if a tiny slit were cut in the loop, so that the circuit is broken?



Solution

Geometry, induced emf, and current

Let the side length of the square loop be a. Let the cross-sectional area of the conducting strip be A. Since the loop is cut from a thick aluminum sheet, A is constant all along the loop.

As the loop falls downward with speed v, the area of the loop that remains inside the field region decreases. Therefore the magnetic flux changes.

If x denotes the vertical overlap of the loop with the field region, then the flux is

$$\Phi_B = B(ax).$$

Hence

$$|\mathcal{E}| = \left|\frac{d\Phi_B}{dt}\right| = Ba\left|\frac{dx}{dt}\right| = Bav.$$

So the induced emf is

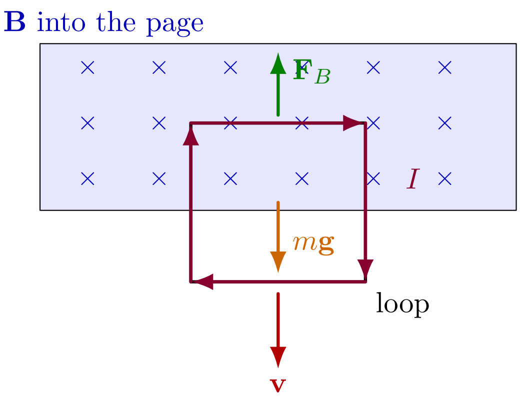
$$\boxed{\mathcal{E} = Bav}$$

in magnitude.

By Lenz's law, the induced current flows so as to oppose the downward motion. Therefore the magnetic force on the top horizontal segment is upward.

The current is

$$I = \frac{\mathcal{E}}{R} = \frac{Bav}{R}.$$



Magnetic force

Only the *top horizontal segment* contributes a net vertical magnetic force:

- the lower horizontal segment is outside the field, so it feels no magnetic force,
- the forces on the two vertical segments are horizontal and cancel each other.

Therefore the upward magnetic force has magnitude

$$F_B = IaB.$$

Substituting $I = \dfrac{Bav}{R}$,

$$F_B = \frac{Bav}{R}\,aB = \frac{B^2a^2}{R}\,v.$$

Thus,

$$\boxed{F_B = \frac{B^2a^2}{R}\,v}$$

Equation of motion

Let the loop move downward. Then

$$m\frac{dv}{dt} = mg - F_B.$$

So

$$m\frac{dv}{dt} = mg - \frac{B^2a^2}{R}v.$$

This is the standard first-order linear equation:

$$\frac{dv}{dt} + \frac{B^2a^2}{mR}v = g.$$

Mass and resistance of the loop

The loop has total conducting length $4a$.

Resistance:

$$R = \rho \frac{4a}{A}.$$

Mass:

$$m = \rho_m (4aA).$$

Now multiply:

$$mR = (\rho_m 4aA)\left(\rho \frac{4a}{A}\right) = 16\rho\rho_m a^2.$$

Hence

$$\frac{B^2 a^2}{mR} = \frac{B^2}{16\rho\rho_m}.$$

This is the key point: **all geometric dimensions cancel out.**

(a) Terminal velocity

At terminal velocity, $\frac{dv}{dt} = 0$, so

$$mg = \frac{B^2 a^2}{R} v_t.$$

Thus

$$v_t = \frac{mgR}{B^2 a^2}.$$

Using the expressions for m and R,

$$v_t = \frac{(\rho_m 4aA) g \left(\rho \frac{4a}{A}\right)}{B^2 a^2}.$$

So

$$\boxed{v_t = \frac{16\rho\rho_m g}{B^2}}$$

(b) Velocity as a function of time

Define

$$\tau = \frac{mR}{B^2 a^2}.$$

From above,

$$\boxed{\tau = \frac{16\rho\rho_m}{B^2}}$$

Then the equation of motion becomes

$$\frac{dv}{dt} + \frac{1}{\tau} v = g.$$

With the initial condition $v(0) = 0$, the solution is

$$\boxed{v(t) = v_t \left(1 - e^{-t/\tau}\right)}$$

where

$$\boxed{v_t = g\tau}$$

So explicitly,

$$\boxed{v(t) = \frac{16\rho\rho_m g}{B^2} \left(1 - e^{-\frac{B^2}{16\rho\rho_m} t}\right)}$$

(c)Time to reach 90% of terminal velocity

Set

$$v(t_{90}) = 0.9\, v_t.$$

Then

$$0.9 = 1 - e^{-t_{90}/\tau}$$

which gives

$$e^{-t_{90}/\tau} = 0.1.$$

Therefore

$$\boxed{t_{90} = \tau \ln 10}$$

(d) Numerical values for aluminum

Take typical values for aluminum:

$$\rho \approx 2.7 \times 10^{-8}\ \Omega \cdot \text{m}, \qquad \rho_m \approx 2.70 \times 10^3\ \text{kg/m}^3, \qquad B = 1\ \text{T}.$$

Then

$$\tau = \frac{16\rho\rho_m}{B^2} = 16(2.7 \times 10^{-8})(2.70 \times 10^3).$$

So

$$\tau \approx 1.17 \times 10^{-3}\ \text{s}.$$

Hence

$$v_t = g\tau \approx 9.81 \times 1.17 \times 10^{-3} \approx 1.14 \times 10^{-2}\ \text{m/s}.$$

Therefore,

$$\boxed{v_t \approx 1.14 \times 10^{-2}\ \text{m/s}}$$

that is,

$$v_t \approx 1.1\ \text{cm/s}.$$

Also,

$$t_{90} = \tau \ln 10 \approx (1.17 \times 10^{-3})(2.3026) \approx 2.69 \times 10^{-3}\ \text{s}.$$

So

$$\boxed{t_{90} \approx 2.7 \times 10^{-3}\ \text{s}}$$

(e) What if the loop is slit (open circuit)?

If a tiny slit is cut in the loop, the circuit is broken, so a sustained current cannot circulate. Therefore:

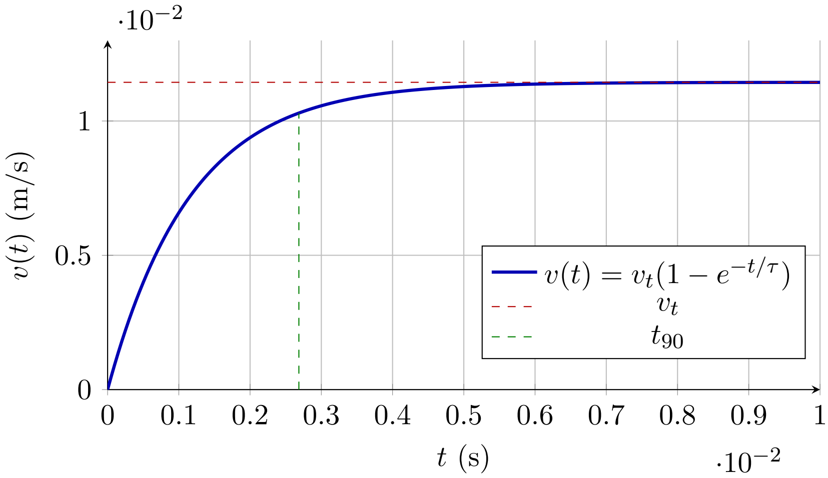
- an emf is still induced in the sense of charge separation,
- but no closed-loop current flows,
- hence there is essentially no magnetic braking force of the kind computed above.

So the loop simply falls freely (neglecting air resistance):

$$\boxed{v(t) = gt}$$

and there is **no magnetic terminal velocity**.

Velocity plot



Side Note: How do we get $v(t)$?

We start from the equation of motion

$$\frac{dv}{dt} + \frac{1}{\tau}v = g.$$

This is a first-order linear differential equation of the form

$$\frac{dv}{dt} + P(t)v = Q(t),$$

where

$$P(t) = \frac{1}{\tau}, \qquad Q(t) = g.$$

The integrating factor is

$$\text{I.F.} = e^{\int P(t)\,dt} = e^{\int \frac{1}{\tau}\,dt} = e^{t/\tau}.$$

Multiplying the differential equation by $e^{t/\tau}$, we get

$$e^{t/\tau}\frac{dv}{dt} + \frac{1}{\tau}e^{t/\tau}v = ge^{t/\tau}.$$

The left-hand side can be written as a single derivative:

$$\frac{d}{dt}\left(ve^{t/\tau}\right) = ge^{t/\tau}.$$

Now integrate both sides with respect to t:

$$ve^{t/\tau} = \int ge^{t/\tau}\,dt = g\tau e^{t/\tau} + C.$$

Therefore,

$$v(t) = g\tau + Ce^{-t/\tau}.$$

Using the initial condition $v(0) = 0$,

$$0 = g\tau + C \quad \Rightarrow \quad C = -g\tau.$$

Hence,

$$v(t) = g\tau\left(1 - e^{-t/\tau}\right).$$

Since the terminal velocity is

$$v_l = g\tau,$$

we finally obtain

$$\boxed{v(t) = v_l\left(1 - e^{-t/\tau}\right)}$$

Physical meaning: At $t = 0$, $v(0) = 0$. As $t \to \infty$, the exponential term vanishes, and therefore

$$v(t) \to v_l.$$

Thus, the velocity gradually approaches the terminal velocity.

Final Results

$$\boxed{v_t = \frac{16\rho\rho_m g}{B^2}}$$

$$\boxed{v(t) = v_t\left(1 - e^{-t/\tau}\right), \qquad \tau = \frac{16\rho\rho_m}{B^2}}$$

$$\boxed{t_{90} = \tau \ln 10}$$

For aluminum with $B = 1$ T:

$$\boxed{v_t \approx 1.14 \times 10^{-2} \text{ m/s}}$$

$$\boxed{\tau \approx 1.17 \times 10^{-3} \text{ s}}$$

$$\boxed{t_{90} \approx 2.7 \times 10^{-3} \text{ s}}$$

If the loop is slit open, then no loop current flows and the loop falls essentially freely:

$$\boxed{v(t) = gt}$$

(with no magnetic terminal speed).

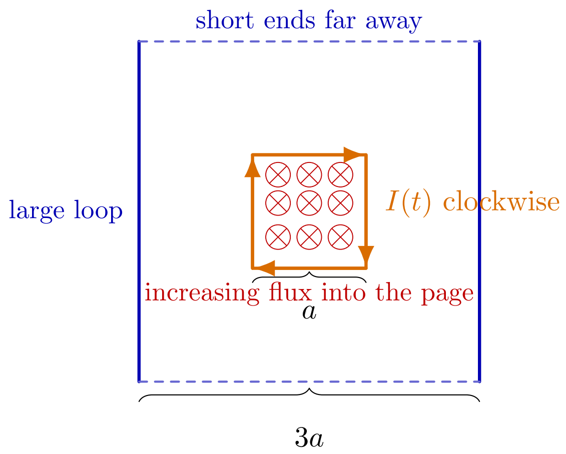
10. A square loop of wire, of side a, lies midway between two long parallel wires, which are $3a$ apart and lie in the same plane as the square loop. The two long wires are the long sides of a much larger rectangular loop; the short ends are so far away that their contribution may be neglected.

A clockwise current I flows in the square loop, and it is increasing at the constant rate

$$\frac{dI}{dt} = k.$$

Find the emf induced in the large loop. Also determine the direction of the induced current in the large loop.

Solution



Strategy

A direct flux calculation through the large loop is possible, but it is much easier to use **mutual inductance reciprocity**:

$$M_{\text{large, small}} = M_{\text{small, large}}.$$

So instead of calculating the flux through the *large* loop due to the current in the *small* loop, we calculate the flux through the *small* loop due to a hypothetical current in the *large* loop.

This is easier because the magnetic field due to two long straight wires is simple.

Important sign convention

We take the clockwise direction of the large loop as the positive reference direction. For a clockwise current in a loop, the associated surface normal is into the page.
Therefore, when we temporarily put a clockwise current I_{big} in the large loop, the left wire carries current upward and the right wire carries current downward. In the region between the two wires, both magnetic fields point

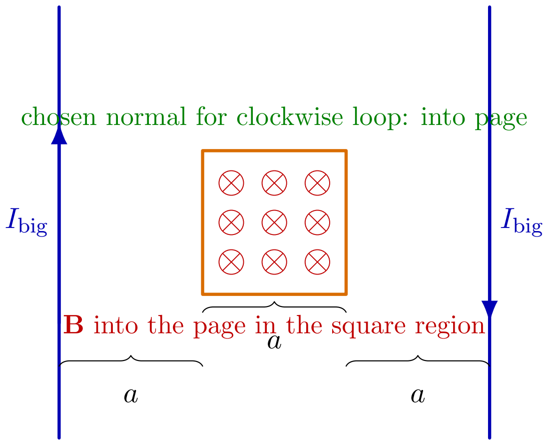
into the page.

Hence, in the figure for the reciprocity calculation, the field symbols should be

$$\otimes,$$

not $\odot$.
If $+\hat{z}$ is chosen out of the page, then the actual B_z is negative. To avoid sign confusion, we calculate the normal component B_n along the clockwise surface normal, which is into the page.

Field of the large loop through the small square



Let the small square occupy

$$-\frac{a}{2} \le x \le \frac{a}{2}, \qquad -\frac{a}{2} \le y \le \frac{a}{2}.$$

The two long wires are located at

$$x = -\frac{3a}{2} \qquad \text{and} \qquad x = +\frac{3a}{2}.$$

For a clockwise current I_{big} in the large loop, the magnetic field between the two wires is into the page. Therefore, the normal component of the magnetic field along the clockwise surface normal is

$$B_n(x) = \frac{\mu_0 I_{\text{big}}}{2\pi} \left[\frac{1}{\frac{3a}{2} + x} + \frac{1}{\frac{3a}{2} - x} \right].$$

Flux through the small square

Since B_n depends only on x,

$$\Phi_{\text{small}} = \int_{-a/2}^{a/2} \int_{-a/2}^{a/2} B_n(x)\, dy\, dx \tag{40}$$

$$= a \int_{-a/2}^{a/2} \frac{\mu_0 I_{\text{big}}}{2\pi} \left[\frac{1}{\frac{3a}{2} + x} + \frac{1}{\frac{3a}{2} - x} \right] dx. \tag{41}$$

Therefore,

$$\Phi_{\text{small}} = \frac{\mu_0 I_{\text{big}} a}{2\pi} \int_{-a/2}^{a/2} \left[\frac{1}{\frac{3a}{2}+x} + \frac{1}{\frac{3a}{2}-x} \right] dx.$$

Now,

$$\int_{-a/2}^{a/2} \frac{dx}{\frac{3a}{2}+x} = \ln 2,$$

and

$$\int_{-a/2}^{a/2} \frac{dx}{\frac{3a}{2}-x} = \ln 2.$$

Hence,

$$\Phi_{\text{small}} = \frac{\mu_0 I_{\text{big}} a}{2\pi} \left(2\ln 2\right).$$

So,

$$\Phi_{\text{small}} = \frac{\mu_0 I_{\text{big}} a}{\pi} \ln 2.$$

Therefore, the mutual inductance is

$$\boxed{M = \frac{\Phi_{\text{small}}}{I_{\text{big}}} = \frac{\mu_0 a}{\pi} \ln 2}$$

By reciprocity, this is also the mutual inductance between the small square loop and the large loop.

Induced emf in the large loop

Using the clockwise direction of the large loop as the positive reference direction,

$$\mathcal{E}_{\text{cw}} = -M\frac{dI}{dt}.$$

Since

$$\frac{dI}{dt} = k,$$

we get

$$\mathcal{E}_{\text{cw}} = -\frac{\mu_0 a}{\pi} \ln 2\ k.$$

Thus,

$$\boxed{\mathcal{E}_{\text{cw}} = -\frac{\mu_0 a}{\pi} \ln 2\ k}$$

The negative sign means that the induced emf is opposite to the assumed clockwise direction. Therefore, the induced current in the large loop is counterclockwise.

The magnitude of the induced emf is

$$\boxed{|\mathcal{E}| = \frac{\mu_0 a}{\pi} \ln 2\ k}$$

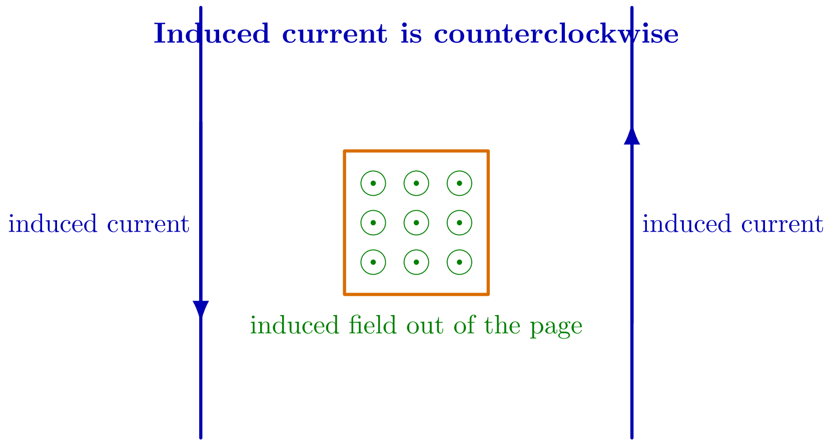
Direction of the induced current

The current in the small loop is clockwise. Therefore, by the right-hand rule, its magnetic field through the central region is into the page.

Since the clockwise current is increasing, the magnetic flux into the page is increasing.

By Lenz's law, the induced current in the large loop must oppose this increase. Therefore, it must produce a magnetic field out of the page in the region between the two long wires.

A counterclockwise current in the large loop produces an out-of-page field between the long wires.



Therefore,

$$\boxed{\text{induced current in the large loop is counterclockwise.}}$$

Final Answer

The magnitude of the induced emf in the large loop is

$$\boxed{|\mathcal{E}| = \frac{\mu_0 a}{\pi} \ln 2 \; k}$$

If clockwise is taken as the positive direction for the large loop, then the signed emf is

$$\boxed{\mathcal{E}_{\text{cw}} = -\frac{\mu_0 a}{\pi} \ln 2 \; k}$$

Hence, the induced current flows in the

$$\boxed{\text{counterclockwise direction.}}$$